

Many different performance metrics exist in networks:

- **Throughput** — how much data gets through per second
- **Jitter** — variation in packet delay
- **Utilisation** — fraction of link capacity in use
- **Delay & packet loss** — how long packets wait and how often they are dropped

Of these, **delay** and **loss** are the most critical in packet-switched networks

Some of the Delay and Loss sources Are:

## 1 Overview — What & Why

### Delay sources:

- Connection setup (TCP handshake)
- Queueing (waiting in buffer)
- Serialization (placing bits on wire)
- Medium access (CSMA, TDMA, ...)
- Retransmissions (lost frames)

### Loss sources:

- Buffer overflow (queue full  $\rightarrow$  drop)
- Collisions (shared medium)
- Checksum/CRC mismatch
- (TCP) session drop

- Unlike propagation or serialisation.
- Queueing delay is *stochastic*:
  - It varies with load and cannot be predicted by a simple formula.
  - This is why we model networks with **queueing theory**.

**Networks** are *packet-switched*: packets share links and wait in buffers  $\Rightarrow$  **delay** and **loss**. We model this with *queueing theory* (Markov chains) to predict  $\mathbb{E}[\text{delay}]$ ,  $P(\text{loss})$ , throughput.

## 2 Kendall Notation — $X/Y/C[/math> $/B]$$

To describe and classify queue types we use **Kendall notation**  $X/Y/C[/math>  $/B]$ , where each symbol describes one aspect of the system:$

- $X$  — Arrival process ( $M$  = Markovian/Poisson)
- $Y$  — Service process ( $M$  = exponential)
- $C$  — Number of servers
- $B$  — Buffer size (omit  $\Rightarrow$  infinite)

### Examples of these are

| Notation | Meaning                            |
|----------|------------------------------------|
| M/M/1    | 1 server, $\infty$ buffer          |
| M/M/K/K  | $K$ servers, $K$ buffer (blocking) |
| M/M/1/B  | 1 server, finite buffer $B$        |

## 3 M/M/1 Queue

The simplest and most fundamental model is **M/M/1**:

- **M** (arrival) — memoryless Poisson process, rate  $\lambda$ ; inter-arrival times are i.i.d. exponential
- **M** (service) — memoryless exponential service times, rate  $\mu$
- **1** — single server; packets served one at a time (FCFS)
- infinite buffer.

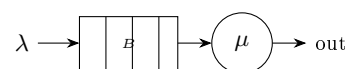
**Arrivals:** Poisson rate  $\lambda$     **Service:** exp. rate  $\mu$     **Utilization:**  $\rho = \lambda/\mu$

Stable only if  $\rho < 1$

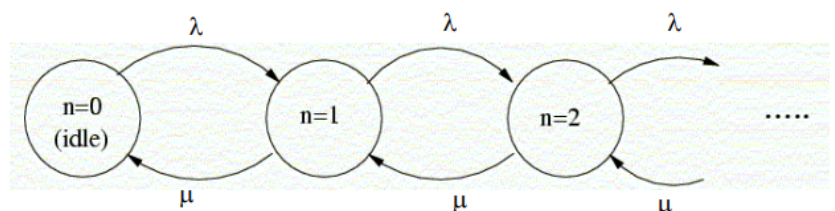
We can represent M/M/1 in two equivalent ways:

- **buffer+server diagram** showing the physical flow
- **birth-death chain** where each state  $n$  is the number of packets currently in the system (waiting + being served).

*Buffer + server* — packets arrive, queue, get served:



*Birth-death chain* — state  $n$  = # packets in system (incl. in service):



### 3b M/M/1 — Steady-State Results

**Steady-state distribution** (geometric):

The probability of finding exactly  $i$  packets in the system:

$$\pi_i = \Pr(Q = i) = (1 - \rho) \rho^i$$

Probability server is idle:  $\pi_0 = 1 - \rho$

**Key metrics** ( $Q = \#$  in system,  $S =$  time in system):

$$\underbrace{\mathbb{E}[Q]}_{\text{avg. queue length}} = \frac{\rho}{1 - \rho}, \quad \underbrace{\mathbb{E}[S]}_{\text{avg. delay}} = \frac{1}{\mu - \lambda}$$

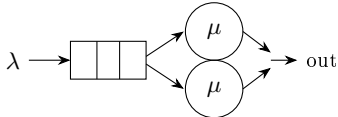
Finite buffer  $B$ :  $P_{\text{loss}} = \rho^B$

**PASTA property:** Poisson arrivals see time averages —  $P(\text{arrival finds } Q \geq B) = \Pr(Q \geq B) = \rho^B$

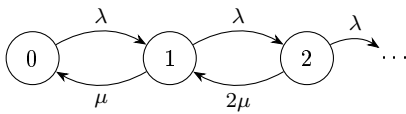
**Extra Stuff (Pure Notes Don't say)**

**Exercise: M/M/2 — two servers, shared queue**

Arrival  $\lambda$  feeds *one* shared buffer; two servers each rate  $\mu$ .



Birth-death — death rate varies by state:



**Note: increasing server speed always better**

Compare three systems, same total capacity  $2\mu$ :

1. M/M/1 rate  $2\mu$  — one fast server
2. M/M/2 rate  $\mu$  each — pooled queue
3.  $2 \times$  M/M/1 rate  $\mu$ , split  $\lambda/2$  each

$$\mathbb{E}[S]_{\text{M/M/1}, 2\mu} < \mathbb{E}[S]_{\text{M/M/2}, \mu} < \mathbb{E}[S]_{2 \times \text{M/M/1}, \mu}$$

Pooling (shared queue) beats splitting.

A single faster server beats pooling.

For M/M/2:  $\rho = \lambda/(2\mu)$ , death rate is  $\mu$  in state 1,  $2\mu$  in states  $\geq 2$ .

### 4 Birth-Death Process (General)

State = packets in system (queue + in service). Transitions only  $\pm 1$ . M/M/1 is the simplest special case ( $\lambda_k = \lambda$ ,  $\mu_k = \mu$ ).

**General steady-state (product form):**

$$\pi_i = \pi_0 \prod_{k=0}^{i-1} \frac{\lambda_k}{\mu_{k+1}}, \quad \pi_0 = \left( 1 + \sum_{i=1}^{\infty} \prod_{k=0}^{i-1} \frac{\lambda_k}{\mu_{k+1}} \right)^{-1}$$

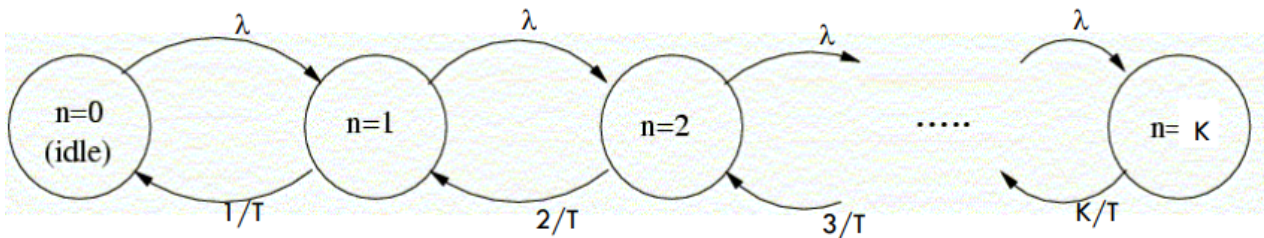
**Flow balance** (global):

$$\mu_i \pi_i = \lambda_{i-1} \pi_{i-1}$$

Rate out of state  $i$  = rate into state  $i$ .

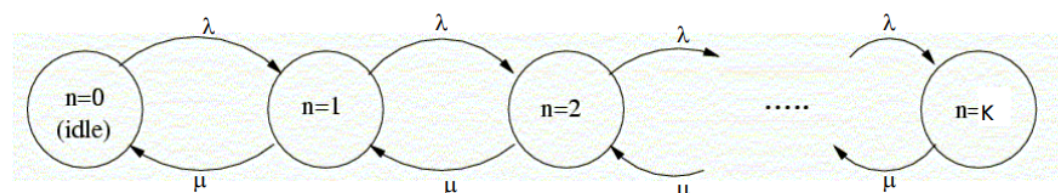
### 5 M/M/K/K Queue — Erlang-B (Circuit Switching)

$K$  channels (servers). Call **blocked** (lost) if all  $K$  busy — no waiting room. Departure rate from state  $i$ :  $i/T$ .



Death rate from state  $i$  is  $i/T$  because each of the  $i$  active calls ends independently at rate  $1/T$ . Blocking probability  $P_B = \pi_K$  — the **Erlang-B** formula.

### 6 M/M/1/K — Finite Buffer



$$\pi_i = \frac{1 - \rho}{1 - \rho^{K+1}} \rho^i \quad (i = 0, \dots, K), \quad P_{\text{loss}} = \pi_K$$

$$\bar{D} = \frac{1}{\lambda(1 - P_{\text{loss}})} \cdot \frac{\rho}{1 - \rho^{K+1}} \left[ \frac{1 - \rho^K}{1 - \rho} - K\rho^K \right]$$